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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice Questions

25 September 2017

1 hour

Additional Material: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

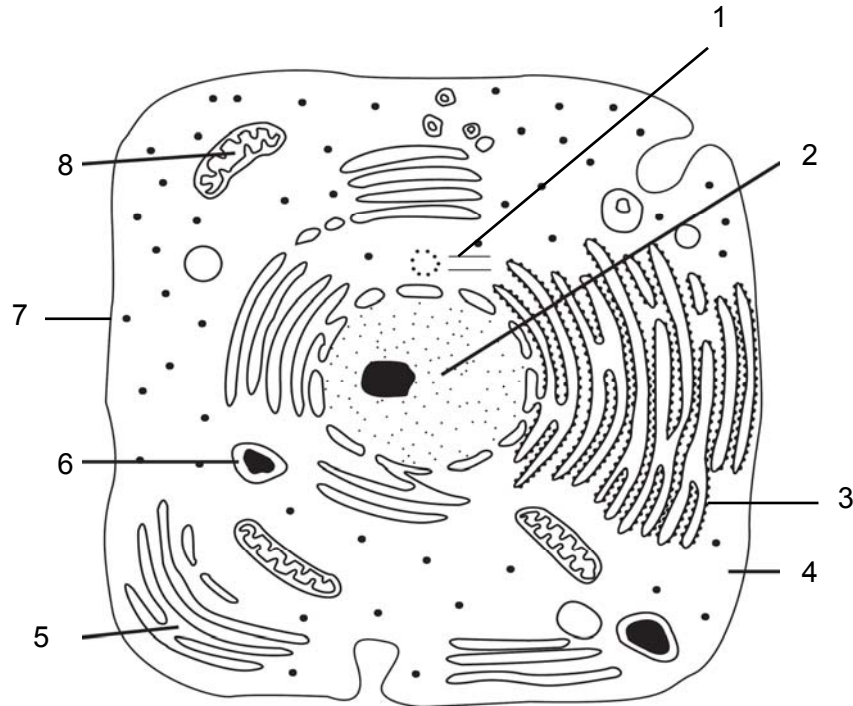
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **22** printed pages and **0** blank page.

Multiple Choice Questions (30 marks)

Answer **all** questions in this section.

- 1 The diagram shows a drawing of an electron micrograph of an animal cell.



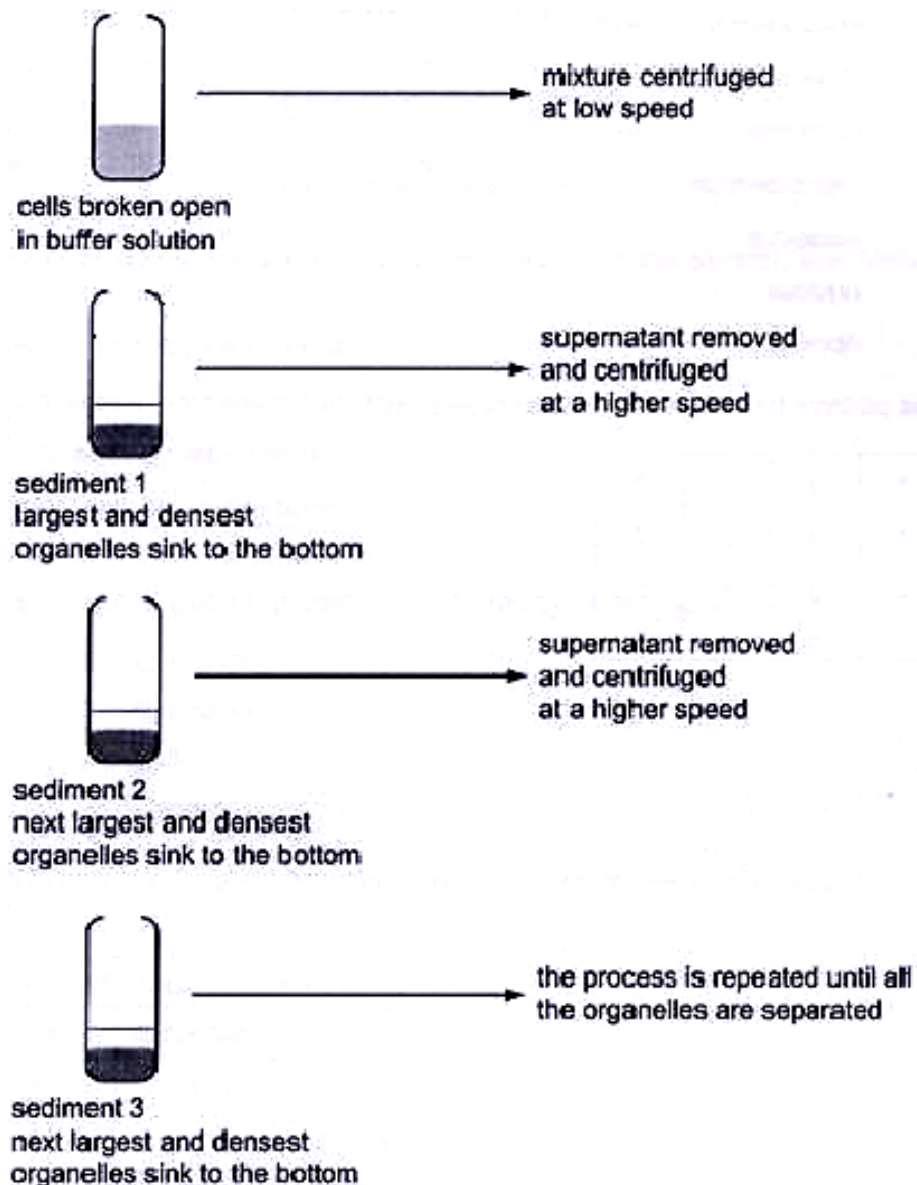
Which of the following describes the corresponding properties of the labelled structures?

	undergoes doubling during cell division	contain enzymes	contains nucleic acids
A	2	6, 8	2, 5, 8
B	2, 4, 8	5, 6, 8	2, 3, 8
C	1, 2, 8	2, 4, 6, 8	1, 2, 7, 8
D	1, 2	2, 3, 4, 5, 6	2, 3, 8

- 2 Which statement is **TRUE** for phospholipids, but not for protein?

- A It has hydrophilic and hydrophobic components.
- B It is synthesized from non-identical sub-units.
- C It can form a barrier to water soluble molecules.
- D It is found in cell membranes.

- 3 Fractionation is a process used to separate cell components according to their size and density. The diagram shows the main stages in fractionation of a plant cell.

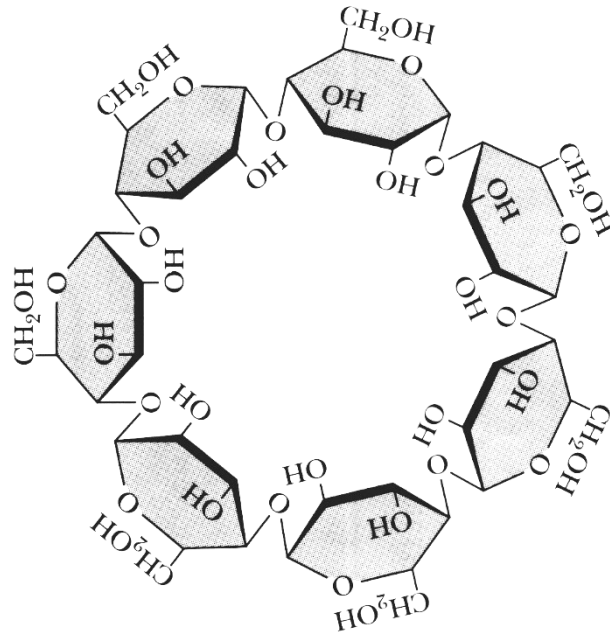


DCPIP and buffer solution (containing glucose, fructose, sodium bicarbonate) were added to each of the sediments, and the mixtures were exposed to light for fifteen minutes. Sediment 2 caused the DCPIP to be reduced.

Which organelle present in Sediment 2 caused reduction of DCPIP?

- A chloroplast
- B mitochondria
- C chloroplast and mitochondria
- D ribosomes

- 4 The diagram shows a circular oligosaccharide molecule.



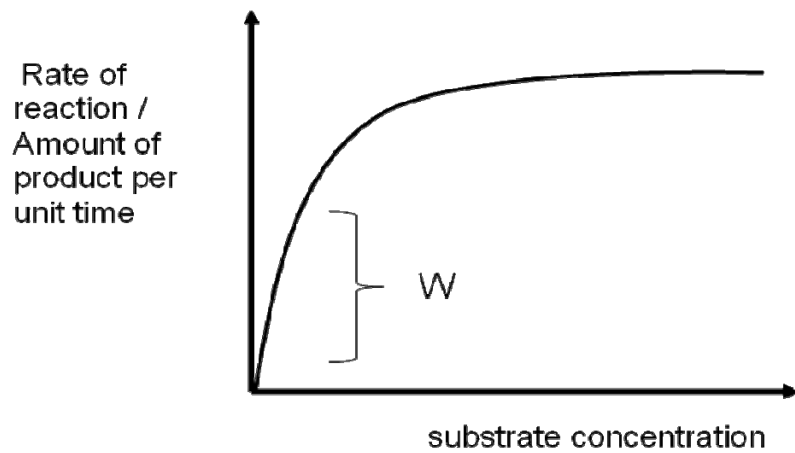
In which other molecule can a similar glycosidic bond be found?

- A lactose
 - B maltose
 - C sucrose
 - D cellulose
- 5 The hydrolysis of triglycerides leads to _____.
- 1 formation of products which are more soluble in water than triglycerides.
 - 2 formation of products which are less soluble in water than triglycerides.
 - 3 an increase in pH.
 - 4 a decrease in pH.

Choose the correct statements to complete the sentence.

- A 1 and 4
- B 2 and 3
- C 2 and 4
- D 1 and 3

- 6 The graph below shows the rate of an enzyme catalyzed reaction occurring in lysosome with increasing substrate concentration. The reaction is carried out at 37°C and a pH of 4 for all substrate concentrations.



Which of the following(s) would result in a decrease in the rate of reaction at W?

- 1 Addition of co-factor
 - 2 Decrease in temperature to 27°C
 - 3 Increase in pH to 9
 - 4 Addition of competitive inhibitor
- A 1 and 4
- B 2 and 3
- C 2, 3 and 4
- D 1, 2, 3 and 4

- 7 Some inhibitors of enzyme reactions bind to the enzyme-substrate complex. Which statements about this type of inhibition are correct?

- 1 The active site changes shape.
- 2 The inhibitor is non-competitive.
- 3 The initial rate of reaction is reduced.
- 4 The maximum rate of reaction (V_{max}) is increased.

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 2, 3, and 4 only

- 8 An insertion mutation occurs in the gene coding for an enzyme, tyrosinase. Nucleotide sequences of the gene (the non-template strand), as well as the corresponding amino acid sequence of tyrosinase, are shown below.

Wild-type allele	ATG	AAG	TTG	GCT	AAA	TGG	GGA
Wild-type protein	Met	Lys	Leu	Ala	Lys	Trp	Gly
Mutant allele	ATG	AAG	TTA	GGC	TAA	ATG	GGG
Mutant protein	Met	Lys	Leu	Gly	-		



Insertion of adenine

Which feature of the genetic code cannot be observed based on the information given?

- A The genetic code is degenerate.
- B The genetic code is punctuated.
- C The code is non-overlapping.
- D The code is universal.

- 9 A 19-base pair long DNA molecule was analysed to find the number of nucleotide bases in each of the polynucleotide strands. Some of the results are shown.

	number of nucleotide bases			
	A	C	G	T
strand 1				4
strand 2		7		5

How many hydrogen bonds are present in this DNA molecule?

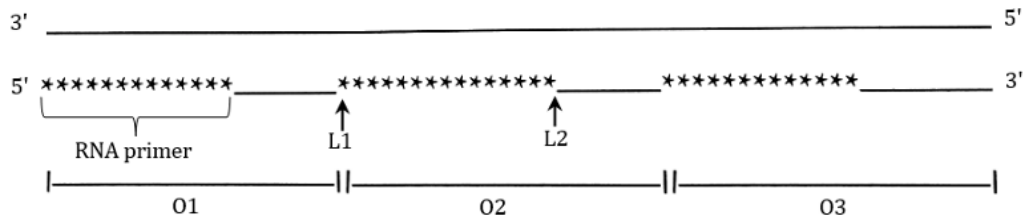
- A 31 B 48 C 39 D 57
- 10 When DNA replicates, new nucleotides containing the common isotope of nitrogen (^{14}N) are used to build new nucleic acids.

In the laboratory, nucleotides can be synthesised using the heavy isotope of nitrogen (^{15}N). Cells grown in ^{14}N nucleotides for many generations are allowed to replicate once using these ^{15}N nucleotides, then twice more using ^{14}N nucleotides.

What will be the percentage of ^{14}N nucleotides in the final molecules?

- A 50% B 75% C 83% D 87.5%

- 11 The diagram shows a DNA template with the lagging strand prior to the removal of the RNA primers.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

	first Okazaki fragment synthesised	site of phosphodiester bond formation catalysed by DNA ligase
A	O1	L1
B	O1	L2
C	O3	L1
D	O3	L2

- 12 The following statements describe various steps in translation.

- 1 Large ribosomal subunit binds to mRNA.
- 2 Small ribosomal subunit binds to mRNA.
- 3 Anticodon of activated tRNA base pairs with codon AUG at the A site.
- 4 Anticodon of activated tRNA base pairs with codon AUG at the P site.

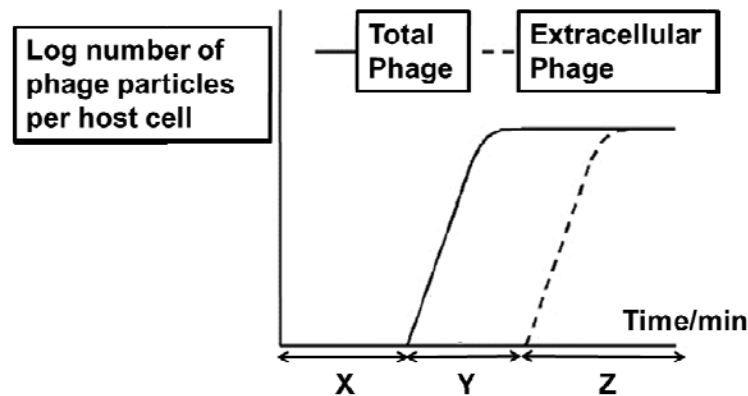
Which of the following statements describe the initiation phase?

- A 1 and 2 only
- B 1, 2 and 3
- C 1, 2 and 4
- D All of the above

- 13** *Pithovirus* was recently discovered and classified as a species of giant virus. It is approximately 1.5 μm in length, larger than the smallest known eukaryotic cell and larger than any known giant virus. It carries double stranded DNA and replicates in the cytoplasm of amoeba, a single cell animal. It carries the genes for transcribing DNA to RNA and genes required for protein synthesis.

Which of the following explains why this organism was classified as a virus?

- A** It is only able to replicate within amoeba.
 - B** It is too large to be known as a eukaryotic cell.
 - C** It carries double stranded DNA, similar to bacteriophages.
 - D** Similar to HIV and influenza virus, it carries enzymes that transcribes its genome.
- 14** The figure below shows a growth cycle of a T4 phage.

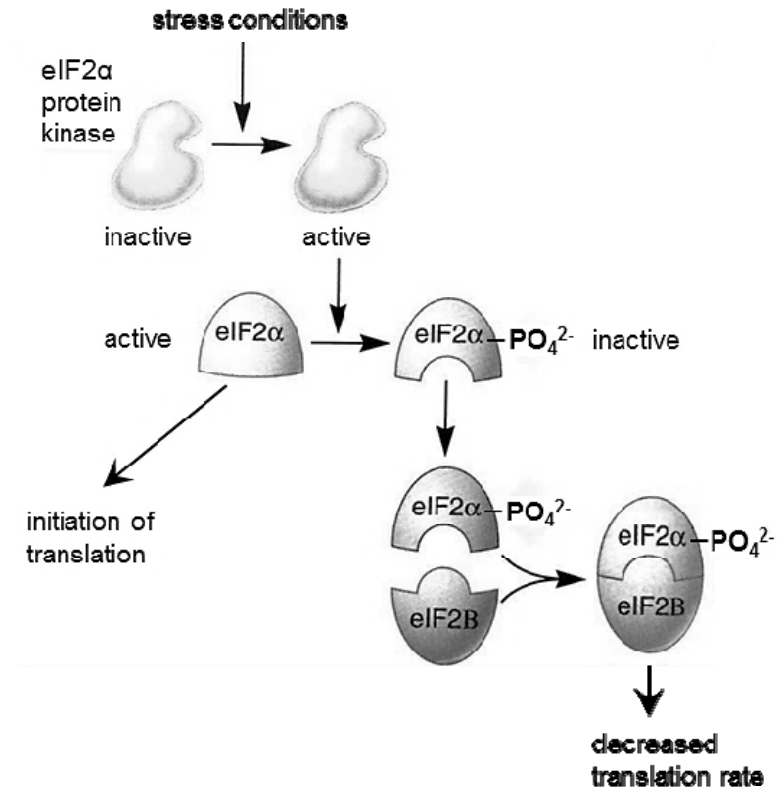


Which of the following statements about **X**, **Y** and **Z** of the growth cycle is correct?

- A** **Y** is the period where there is just active viral DNA replication and protein production.
- B** **X** is the eclipse period where the phage just infected the host cell.
- C** **X** corresponds to the period where the phage exists as a prophage.
- D** Period **Z** will correspond to the death of host cells.

- 15** When trypsin converts chymotrypsinogen to chymotrypsin, some molecules of chymotrypsin bind to a repressor, which in turn binds to the operator and prevents further transcription of trypsin gene. This is most similar to which of the following operons?
- A** trp operon during lack of tryptophan
 - B** trp operon during abundance of tryptophan
 - C** lac operon during lack of lactose
 - D** lac operon during abundance of lactose

- 16 The diagram shows a mechanism by which gene expression is controlled during translation.



Which statements are correct?

- 1 Phosphorylation by eIF2α protein kinase changes the conformation of and activates eIF2α.
- 2 The concentration of eIF2B affects the rate of translation by inhibiting translation initiation.
- 3 This regulation results in decreased rate of mRNA translation under stress conditions.
- 4 The eIF2α-eIF2B complex is recognised by proteasomes for selective degradation due to the presence of the phosphate group.

- A 1 and 2
 B 1 and 4
 C 2 and 3
 D 3 and 4

17 The table compares the genomes of various organisms.

organism	classification	number of chromosomes	size of genome / Million base pairs	approximate number of protein-coding genes
<i>Haemophilus influenzae</i>	bacteria	1	1.8	1700
<i>Saccharomyces cerevisiae</i> (yeast)	eukarya	16	12.1	5900
<i>Drosophila melanogaster</i> (fruit fly)	eukarya	4	180	13000
<i>Oryza sativa</i> (rice)	eukarya	12	440	50000
<i>Canis familiaris</i> (dog)	eukarya	39	2400	19000
<i>Homo sapiens</i> (human)	eukarya	23	3000	19000

Which statement can be inferred from the table?

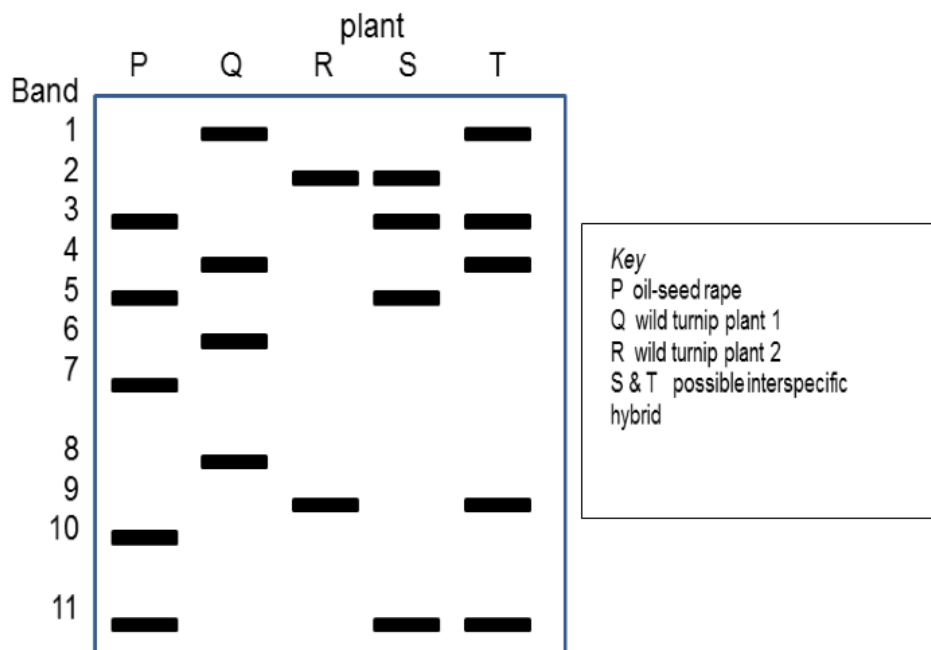
- A On average, each gene in *Haemophilus influenzae* is about 1000 base pairs.
- B On average, each gene in *Saccharomyces cerevisiae* is about 2000 base pairs.
- C The more chromosomes there are in an organism, the larger the genome size.
- D The difference in the number of genes between *Canis familiaris* and *Homo sapiens* is due to the difference in size of genome.

- 18** Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from these plants and their DNA analysed using gel electrophoresis.

Plants S and T were suspected to be interspecies hybrids of oil-seed rape (P), and wild turnip plant 1 (Q) or wild turnip plant 2 (R).

Using the key provided below, determine

- (i) whether plants S and/or T are interspecific hybrids and
- (ii) the parental plants they are derived from



	(i) Interspecific hybrid	(ii) Parent plants
A	Both S and T	S: P x R T: P x Q
B	Both S and T	S: P x Q T: P x R
C	T only	Q x R
D	S only	P x R

- 19** Yeast cells without a *cdc25* gene cannot divide. This gene is active throughout the cell cycle, steadily building up the concentration of a protein, p80cdc25. This protein activates a kinase which regulates other proteins involved in cell division, but does not seem to affect other cell processes. When the p80cdc25 protein reaches a critical concentration, mitosis starts.

Which changes will be seen if p80cdc25 is produced at a faster rate than usual?

- 1 faster cell cycle
- 2 slower cell cycle
- 3 smaller cells
- 4 larger cells

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

- 20** Observations of cancer development show the following.

- Control genes code for the synthesis of proteins that act at different points of the cell cycle to promote or block its completion.
- There are a number of genes involved in the control of cell division.
- The risk of cancer developing increases with age.
- Cancer cells no longer respond to signals that regulate cell division and growth of most cells.

Which statement could explain why cancer occurs?

- A** Mutated alleles of control genes produce proteins that inhibit the proteins produced by the normal alleles of these genes.
- B** Mutation of an allele of any of the genes involved in the cell cycle allows faulty cells to be replicated leading to a tumour.
- C** Mutations of control genes that accumulate in a cell over time slow down the cell cycle.
- D** Mutations of genes that code for cell surface receptors prevent the cell from receiving signals that stimulate cell division.

- 21** Purple buds of the morning glory flower, *Ipomoea*, open into blue flowers. As the flower opens, the pH on the vacuoles of the flower epidermal cells increases and this results in a change of colour from purple to blue.

A mutant purple-flowered morning glory plant carries recessive alleles of a gene **B/b**, coding for a membrane-bound ion pump, and is unable to increase the pH of the vacuole.

Both normal blue flowers and mutant purple flowers have the same anthocyanin pigment, coded by the dominant allele of the gene **A/a**. Plants with **aa** cannot produce anthocyanin and they have white flowers.

The genes **A/a** and **B/b** are on different chromosomes.

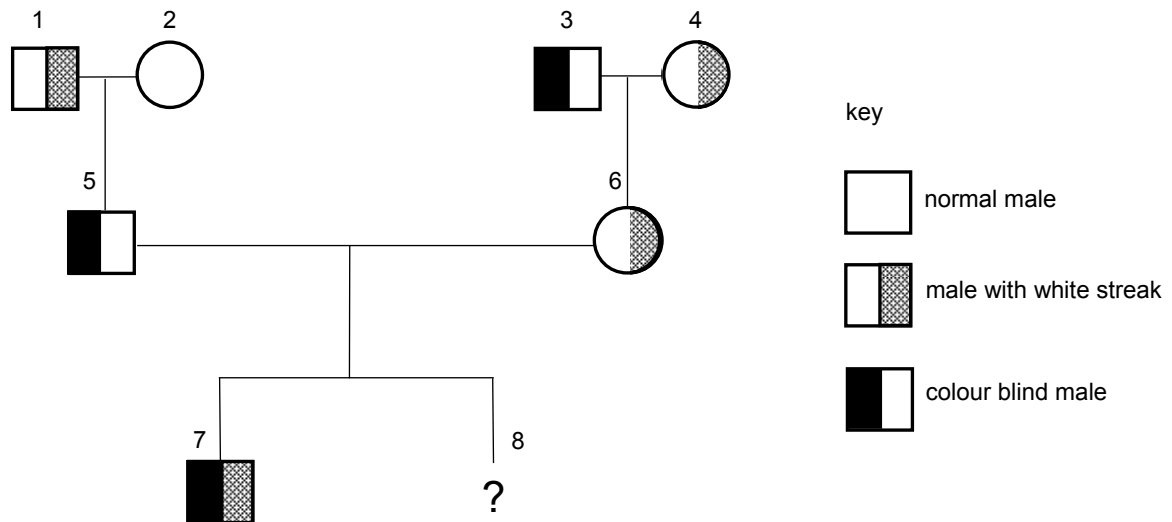
A blue-flowered morning glory plant was crossed with a purple-flowered plant. Their offspring consisted of plants which are blue-flowered, purple-flowered as well as white-flowered.

What were the genotypes of the blue-flowered and purple-flowered parents?

	Blue-flowered parent	Purple-flowered parent
A	AABB	AaBb
B	AaBb	Aabb
C	AaBB	Aabb
D	AABb	aabb

- 22** Colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, X^b , is recessive to the allele for normal colour vision, X^B . The gene controlling the presence of a white streak in the hair is not sex-linked, with the allele for the presence of a white streak, H , being dominant to the allele for the absence of a white streak, h .

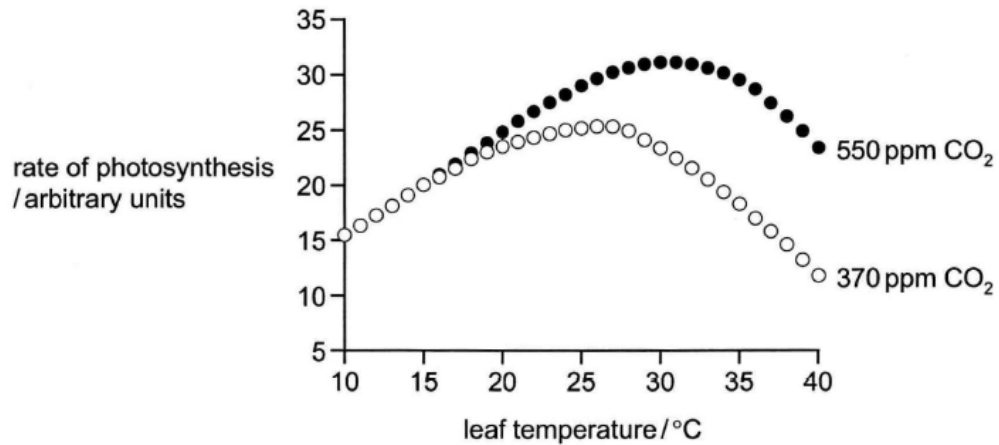
The diagram shows a pedigree in which some of the individuals have colour blindness or have a white streak present in the hair.



What is the probability that individual 8 is a male with the same phenotype as individual 7?

- A** 0.125
- B** 0.25
- C** 0.5
- D** 0.75

- 23** The graph shows the results of increased concentrations of carbon dioxide on soy bean photosynthesis at various leaf temperatures. Carbon dioxide concentration is measured in ppm (parts per million). Light intensity was at an optimum level.



Which conclusion concerning the data in the graph is valid?

- A** At all temperatures up to 15°C, carbon dioxide concentration is limiting. Above 15°C, temperature becomes the limiting factor.
- B** Supplementing plants with carbon dioxide is only effective at temperatures above 25°C.
- C** The photosynthetic rate obtained at the optimum temperature for 370ppm CO₂ could be achieved at a temperature 5°C lower using an increased concentration of CO₂.
- D** When light intensity and temperature are limiting, increased carbon dioxide concentration increases the rate of photosynthesis.

- 24** Some apples can be stored in controlled atmospheric conditions for up to a year. Taste and texture are maintained by using conditions that reduce the production of a fruit-ripening plant hormone while limiting the build-up of ethanol. Ethanol damages the fruit.

The storage conditions needed include low temperature (1°C), high carbon dioxide concentration (1.2%) and low oxygen concentration (0.9%).

Why are these conditions needed?

- 1 Low oxygen concentration favours anaerobic respiration.
- 2 Enzyme activity is reduced.
- 3 Conversion of sugar to ethanol is minimised.
- 4 High carbon dioxide concentration promotes photosynthesis.

A 1, 2 and 3

B 1, 2 and 4

C 2 and 3 only

D 3 and 4 only

- 25** The concentration of second messenger is regulated during cell signalling process.

Which of the following statements support cAMP as an effective second messenger?

- 1 Activated adenylyl cyclase converts ATP to cAMP.
- 2 Adenylyl cyclase synthesizes many cAMP molecules.
- 3 cAMP is soluble in the nucleus.
- 4 Phosphodiesterase breakdown cAMP to AMP.

A 2 and 3 only

B 1 and 2 only

C 1, 2 and 4 only

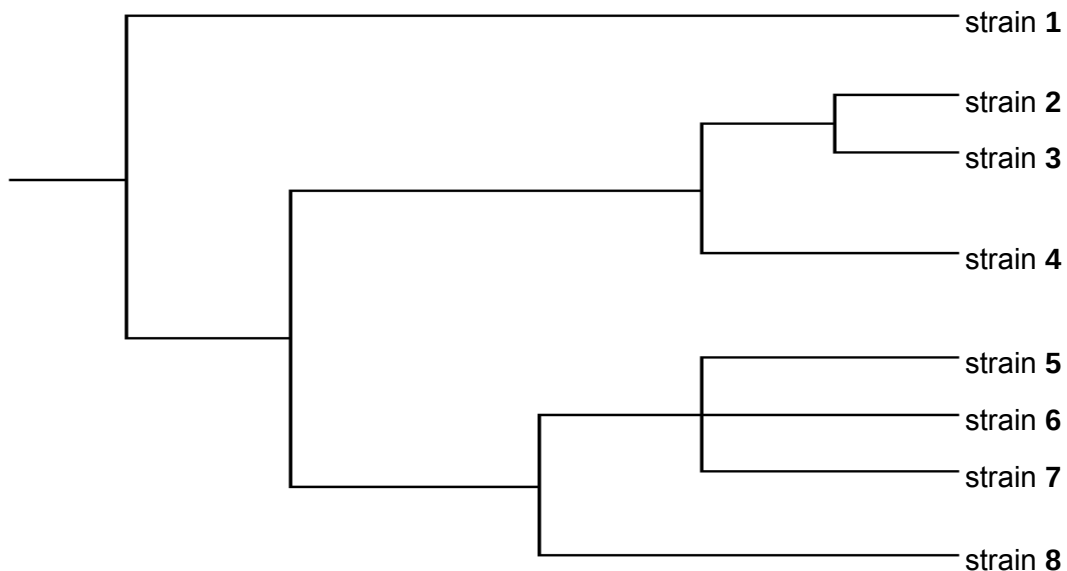
D 1, 3 and 4 only

- 26** In humans, Severe Acute Respiratory Syndrome (SARS) is a serious form of pneumonia. SARS is caused by a coronavirus that was first identified in 2003.

Scientists suspected that the virus had been transmitted to humans from some other animal. Testing was completed on several animal species. Strains of the coronavirus similar to those found in humans were identified in different species of horseshoe bats (genus *Rhinolophus*) and palm civets (*Paguma larvata*). Samples were taken from the different sources and the virus's RNA from each sample was sequenced.

The molecular information enabled the scientists to draw an evolutionary tree for different strains of the coronavirus.

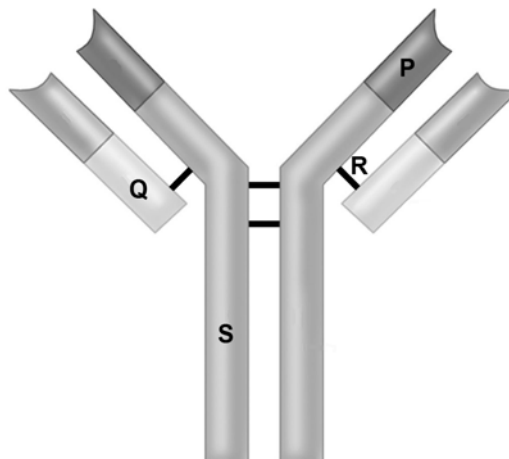
The following evolutionary tree was drawn. Strain 7 is found in palm civets, and strains 5 and 6 in humans. All other strains are found in different species of horseshoe bats.



Which suggestion could provide an explanation for the evolution of strains 5, 6 and 7?

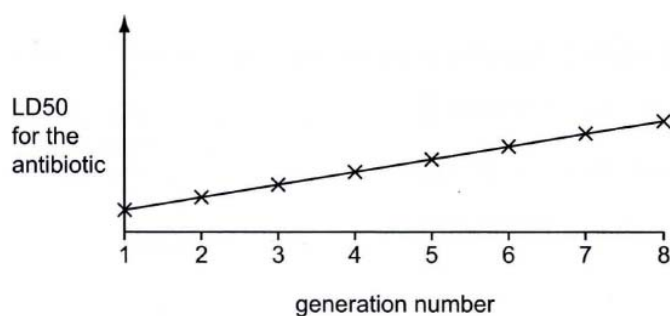
- A** Strains 5, 6 and 7 share the most recent common ancestor, suggesting that strain 7 evolved into strains 5 and 6 over time.
- B** Strains 5, 6 and 7 belong to the same genus and species but different families.
- C** Strains 5, 6 and 7 possess a shared derived character that distinguishes them from strain 8.
- D** Strain 1 is the most distant common ancestor of strains 2 to 8 as it underwent allopatric speciation in the process of evolution.

27 The diagram below shows an antibody.



Which statement correctly match the structure to its function?

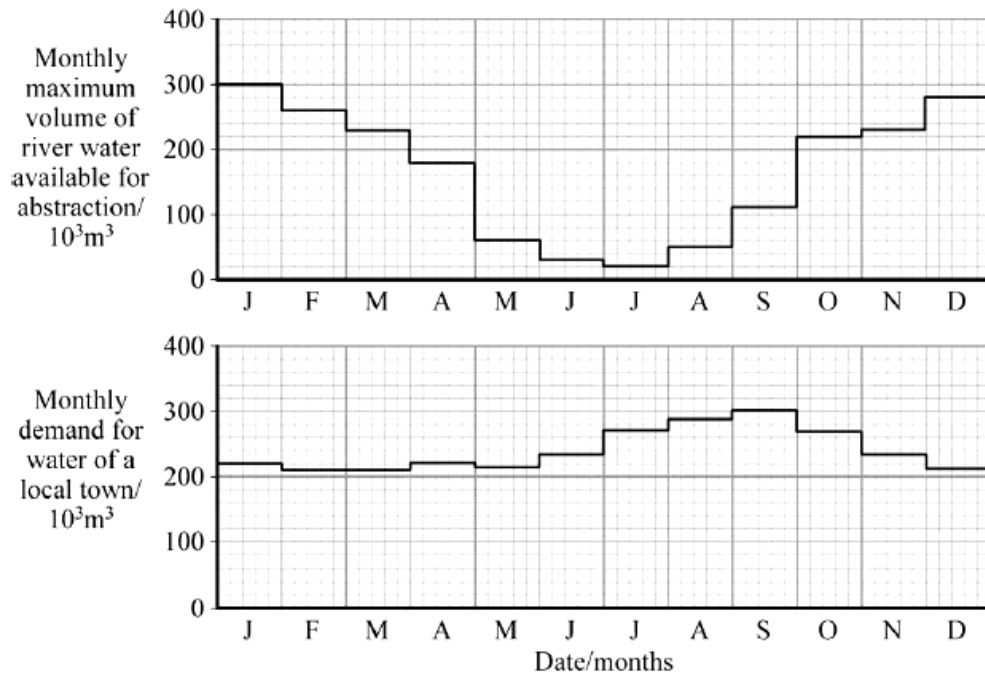
- A Structure **P** has a unique antigen binding site that recognise the antigen on the bacteria.
 - B Structure **Q** is the constant region which binds to receptors on the macrophages and induce phagocytosis of the bacteria.
 - C Structure **R** ensures flexibility of the antibody to coat surface of the bacteria.
 - D Structure **S** is the variable region that determines the isotype of the antibody.
- 28 LD50 (lethal dose 50) is the concentration of a substance that kills 50% of a population. The LD50 was assessed in a population of bacteria constantly exposed to an antibiotic.



What explains the results shown in the graph?

- A Sudden mutations occur in all bacteria in the population.
- B The bacteria develop an enzyme to destroy the antibiotic.
- C The bacteria develop metabolic pathways to destroy the antibiotic.
- D Bacteria that survive pass on genes for resistance to the next generation.

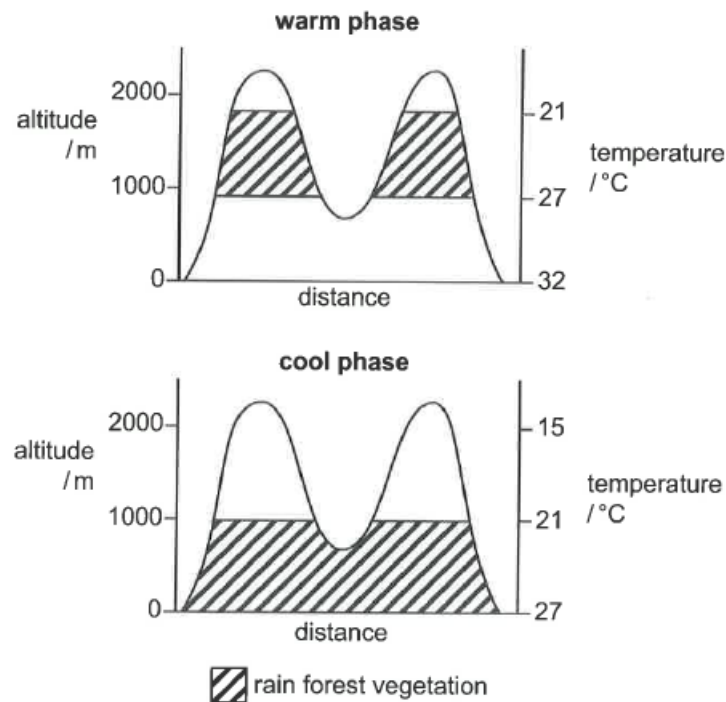
- 29** A small town depends on a river for water supply. The graph shows the projected water availability from the river and the demand for water in the town. The winter months are from July to September and the summer months are from January to April.



Which of the following cannot be concluded from the information given?

- A** The town faces severe water stress in the months of June and July.
- B** Climate change will increase the water stress on the town by causing the winter ice to melt more rapidly than normal in December and January.
- C** The increase in temperature will result in more precipitation to occur as rain rather than snow thus the town will suffer from low water supplies by summer's end.
- D** The town faces no water shortage as it is natural for the river to change its volume and water level due to the seasons.

- 30 The diagram shows the topographical profile of two mountains in the tropics during natural warm and cool phases in the Earth's climate. The shape of the lines corresponds to a vertical section through the mountains to show their height and shape. The distribution of rain forest vegetation is also shown.



Which of the following statement is correct?

- A The rain forest vegetation moves to stay in the suitable range of temperatures of 21 to 27°C.
- B The lower altitudes are too hot during the warm phase in the Earth's climate for the growth of the rain forest vegetation.
- C Climate change causes the rain forest vegetation distribution to increase in altitude when the cool phase changes to the warm phase.
- D The changing rain forest vegetation distribution decreases evolution as selection pressure remains the same.

END OF PAPER

2017 Y6 Preliminary Exam H2
MCQ Answer Scheme

1	D	16	C
2	C	17	A
3	C	18	D
4	B	19	A
5	A	20	A
6	C	21	B
7	C	22	A
8	D	23	C
9	B	24	C
10	D	25	C
11	D	26	C
12	C	27	C
13	A	28	D
14	D	29	D
15	B	30	B